

Gan Minghao

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EDUCATION

The University of Hong Kong <i>Exchange Student, School of Computing and Data Science</i>	Jan. 2026 – Present Hong Kong
Sun Yat-sen University <i>School of Software Engineering</i>	Sep. 2023 – Present Guangzhou/Zhuhai, China
– Academic Records: GPA 4.2/5.0, Rank: 10/231 (Top 5%)	

SCHOLARSHIP & AWARDS

National Scholarship for Undergraduates , Ministry of Education of China	2023
First Prize, Outstanding Student Scholarship , Sun Yat-sen University	2023
Special Scholarship , School of Software Engineering, Sun Yat-sen University	2023
Second Prize, Outstanding Student Scholarship , Sun Yat-sen University	2024
Fung Scholarship , Fung Foundation & The University of Hong Kong	2025

RESEARCH EXPERIENCE

NAISS Lab, The University of Hong Kong <i>Research Intern (Supervised by Prof. Shinan Liu)</i>	Hong Kong Jan. 2026 – Present
– Agent Infra Exploration – KV Cache Reuse across Architectures and Positions. Investigated cross-architecture and positional KV cache reuse strategies to reduce redundant prefill computation in multi-turn agent interactions. – Agentic Serving Scheduling. Explored scheduling optimizations targeting KV cache idle periods from tool-call latency to improve serving throughput. – Phase-aware KV Cache Quantization. Analyzing quantization-induced accuracy loss in long-context agent scenarios across different execution phases.	
InPlus Lab, Sun Yat-sen University <i>Research Intern (Supervised by Prof. Wuhui Chen)</i>	Guangzhou, China Nov. 2024 – Dec. 2025
– Project: Low Latency LLM inference system on consumer-grade GPUs via Activation Sparsity – Designed a bottleneck-aware neuron caching policy to maximize GPU workload in sparse computation. – Optimized a hybrid computing pipeline enabling asynchronous GPU&CPU execution with on-the-fly weights I/O. – Developed custom CUDA kernels for sparse matrix multiplication, optimizing hardware parallelism. – Achieved up to $3.7\times$ speedup in inference throughput over state-of-the-art systems; work accepted by OSDI 2026.	

PUBLICATIONS

Kairox: Adaptive GPU-CPU Hybrid LLM Inference via Online Neuron Balancing	OSDI 2026
Yapeng Jiang, <u>Minghao Gan</u> , Zicong Hong, Wuhui Chen, Junyuan Liang, Yue Yu, Meng Guo, Zibin Zheng	

SKILLS

Programming : C/C++, CUDA, Python
Frameworks : Llama.cpp, GGML, vLLM, PyTorch
Tools : Git, SSH, Docker, Linux, LaTeX

LANGUAGE PROFICIENCY

IELTS Academic : 7.5 (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.5)